Giovanni Tauro

R&D Engineer, PhD Candidate| Embedded Systems & IoT



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LinkedIn GitHub Scholar Portfolio GrabCAD

R&D Engineer and PhD Candidate with 5+ years at CNR-STIIMA building embedded and IoT systems for health research. For OmniTrack I wrote the C++ firmware of ESP32 wearable devices (UWB, IMU, BLE) and integrated them with a ROS 2 system on Linux, from first board bring-up to validation with users.

Work Experience

R&D Product Engineer, STIIMA, Lecco
Feb 2021 – Present

- Delivered 4+ functional prototypes across embedded electronics, wearable devices, and interactive systems, from concept to technology demonstration.
- Defined the architecture of multimodal IoT platforms, integrating wearable sensing, indoor localization, BLE communication, and data logging.
- Selected components and tested hardware and communication protocols (ESP32 UWB, BLE, MQTT) for a distributed IoT feedback system in the V.I.T.A. project.
- Defined hardware/software requirements with clinicians and deployed PhiCube devices in Italian clinics within Fit4MedRob (MUR, national programme).

`C++ • Python • ESP32 • Arduino • PlatformIO • BLE GATT • MQTT • ROS2 • Linux • Git`

Head of Product (Co-Founder), Rehabilia Technologies s.r.l., Milan
May 2023 – Present

- Lead product development and prioritisation with engineering, design, and clinical stakeholders, from user research to validation.
- Validate product decisions with usability testing and study data: PhiCube reached SUS 85.8/100 with 19 children and is in two ongoing multicentric studies with 134 patients.

Selected Projects

GIT_1
GIT_2
GIT_3

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

- Wrote the C++ firmware of the wearable tags and anchors (ESP32-S3, PlatformIO, FreeRTOS tasks): UWB ranging on DW1000/DW3000 over SPI, BNO055 IMU over I2C, I2S audio feedback, and battery monitoring.
- Connected the tags to a ROS 2 system on Linux through micro-ROS, with a custom BLE transport next to serial; ROS 2 nodes in Python estimate position by multilateration and log each session in MongoDB.
- Brought up and debugged the boards on the bench and in the field, up to a study with 16 blind participants.

`C++ • ESP32-S3 • FreeRTOS • micro-ROS • ROS2 • BLE GATT • Python • Linux`

PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

WEB

- Designed a modular rehabilitation device combining UX research, mechanical design, embedded systems, and multimodal interaction.
- Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.

`UX • Interaction Design • CAD • Prototyping • Assembly • Usability`

RehaBalloon: Interactive Platform

GIT

- Prototyped a low-cost rehabilitation device in 24 hours during the LMU Munich "Hacking with Hardware" challenge, integrating pressure sensing, visual feedback, and embedded electronics.
- Ran a TensorFlow Lite classifier on an Arduino board to recognise exercise patterns and record rehabilitation activity metrics in real time.

`Seeed Studio • Sensors • C++ • TensorFlow • Machine Learning`

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

PhD Candidate, expected 2027. Topic: IoT-based Multimodal Navigation System – OmniTrack

- Research Mobility, Utrecht University, NLD: Sensory navigation and experimental psychology.
- Research Mobility, Toyo University, JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design & Engineering** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB

- Research Mobility, Chiba University, JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Skills

Embedded ESP32, Arduino, PlatformIO, FreeRTOS, micro-ROS, UWB, IMU, audio and haptic feedback

Protocols BLE GATT, MQTT, UART, SPI, I2C, I2S

Code C++, Python, C#, JavaScript, ROS 2, Linux, Git

Certifications

2026 **Badge — IoT and Digital Transformation** *Cisco, Online*

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*

2024 **.ppt — Robotics and Allied Technologies for Rehab** *MUR, Rome, Italy*

Patent

2023 Malosio, M., Nicora, M. L., Redaelli, D., Tauro, G. WO2023214349A1 *Device for bilateral rehabilitation*.

Languages

Italian: Native English: Advanced (IELTS 2017) French: Basic (A2)

Lecco, October 13, 2026



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